



Tracker 2026 Data Taking Anomaly

Faser General Meeting

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on behalf of Tracker Team

University of Liverpool

11 September 2026



Particle Physics

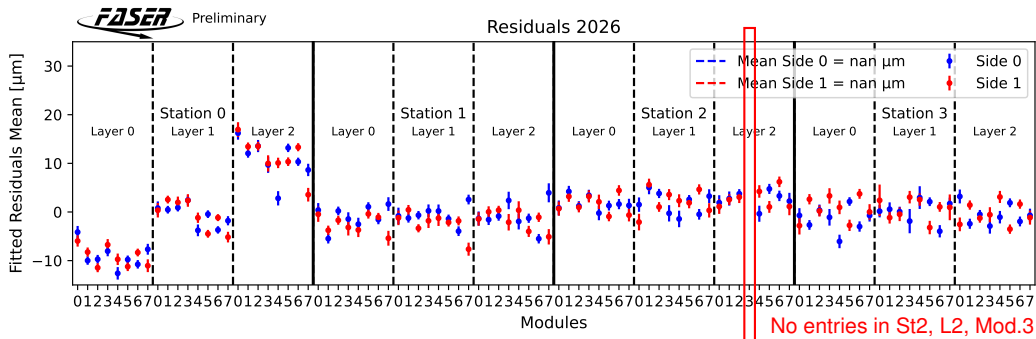
Introduction

- ❖ The Tracker has been taking data since 2022 and has been doing it quite well.
- ❖ We see some potential issues in the 2026 data taking specifically in a specific module.
- ❖ This presentation will discuss this issue.
- ❖ We are quite early in understanding this.
- ❖ But thought it was best to make everyone aware of the situation.
- ❖ We will discuss the issue in a more “chronological” perspective rather than a diagnostic one.

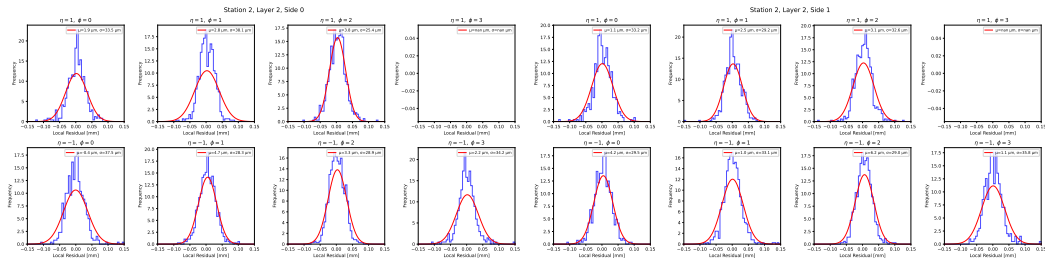
Quick Recap on Tracker

- ❖ TODO: Add tracker diagram and explain the stations and layers.
- ❖ Explain numbering of modules...

- ❖ First seen by Haruhi when trying to calculate unbaised residuals for 2026
- ❖ See Tracker Meeting Aug 12, 2026.

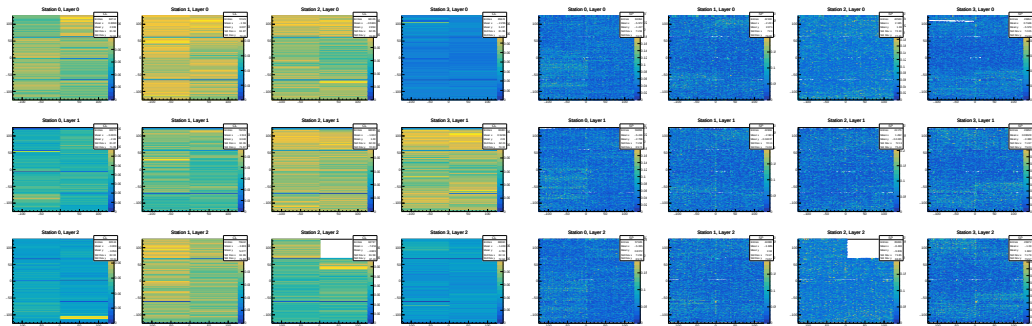


- Initially suspected to be a fit failure/stats issue.



- Quickly ruled out as nearby modules (in same station) had entries.

Looking at Clusters/Space Points



(a) Clusters from Run #022300

(b) Space Points from Run #022300

Figure: SCT Clusters and Space Points from Run #022300. Note that these are subset of the space points that are from the same layer.

Similar Story with SCT Hits itself

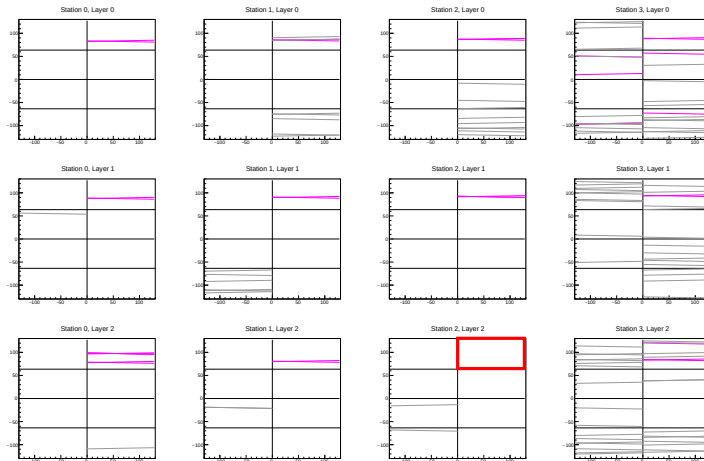


Figure: SCT Hits from Event #2840829, Run #022300. Hits missing in the highlighted module.
Note that out of time HITS (!01X) are shown in grey.

What does the Track look like?

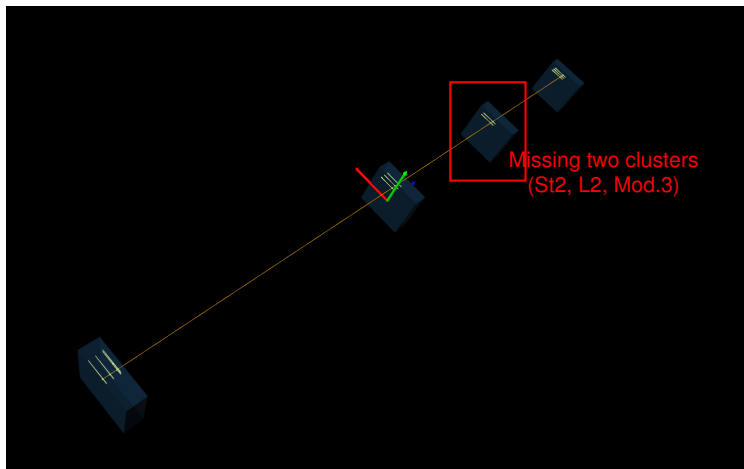


Figure: Event #2840829, Run #022300

What could it mean for the Tracking?

- ❖ It appears as though one module does not have any data
 - ❖ This is probably not completely true
- ❖ In theory this should just be a reduction of two measurements in the final track
- ❖
 - ❖ The Track- χ^2 is lower for these events (Add chi2 plots to backup)
 - ❖ But overall quality (χ^2/NDF) is still good.
 - ❖ Impact on fiducial volume is around X% (See slides from Oscar in backup)
- ❖ More importantly, does the missing hits cause issues with track reconstruction?
 - ❖ This is a harder question to answer.
 - ❖ Needs a more detailed study to quantify this.
 - ❖ Ideally, shouldn't since the seeding should still pick up everything needed for the track.

Cluster Per Run

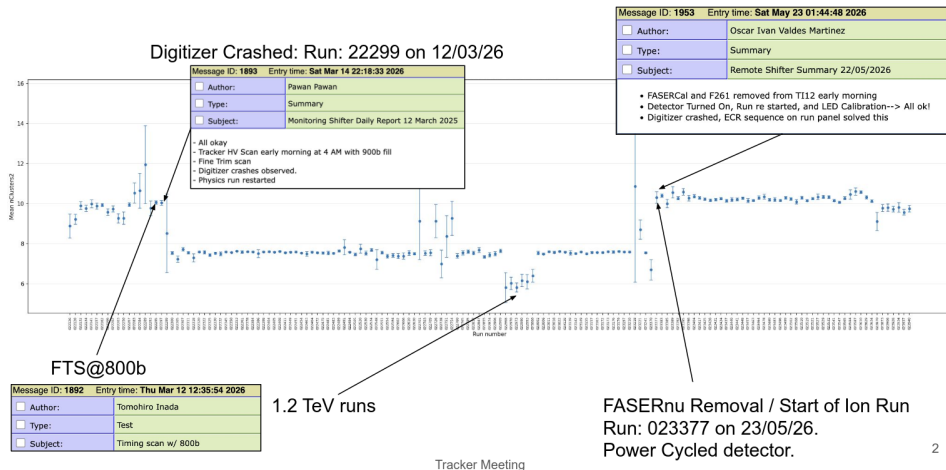


Figure: Timeline of events from Oscar see Tracker Meeting Aug 25 2026

Timeline Summarized

- ❖ Thursday Mar 12 2025: Fine Time Scan, after HV Scan
- ❖ Thursday Mar 12 2025: Fine time adjustment
- ❖ March 13-14 2025: Digitizer Crashes being observed
- ❖

Tangent: Possible Timing Issues?

Caveat: How do we read the raw data?

- ❖ We saw that the SCT Raw Data was broken
- ❖ All of the checks so far uses Calypso
 - ❖ Brain pointed out that this implicitly assumes that Calypso decoded the data correctly.
- ❖ This opens up new questions
 - ❖ Is the raw data actually broken? If so, why?
 - ❖ If not, why is Calypso not decoding it correctly?
 - ❖ Why isn't Calypso flagging an error?

Reconstruction

❖ Reconstruction does flag an error...But

ToolSvc.TrackerDataDecoderTool

ERROR Online module 0 reports one or more errors.

❖ This specific error is flagged from here TrackerDataDecoderTool.cxx#L174

```
else if (sctEvent->HasError())
{
    // FIXME: Should we return failure here?
    ATH_MSG_ERROR("Online module " << onlineModuleID << " reports one or more errors.");
    continue;
}
```

- ❖ While we report the error, we don't handle it in anyway, we “continue” over it.
- ❖ This is partly why we didn't catch it earlier in the prompt reconstruction.
 - ❖ this particular module has an error flag set in the raw data.
 - ❖ we don't consider this a failure and we don't load the data for the module.
 - ❖ other modules in the same event are still loaded and processed.
 - ❖ and we still continue the reconstruction albeit with “missing” data.

End Result

- ❖ We have a valid “event” once Calypso is done reading the raw data.
 - ❖ This event does not really have any data from the specific module.
 - ❖ If the track did not pass through the module, then it is not a problem.
 - ❖ But if it did, then we don't use the specific SCT HITs.
- ❖ Only way to stumble on this would be
 - ❖ Either by looking for ERRORS in the log
 - Note that this isn't guaranteed due to messaging limits (see `MessageSvc.cpp#L333`)
 - Do need to check if this is the case here.
 - ❖ Or by specifically looking for the tracks that pass through this module
 - this we do not usually do
 - we briefly did this for 2024 data (see Track DQ 2024)
- ❖ Ideally if we have an error we should stop and handle it

- ❖ The raw data dump has no errors nominally
 - ❖ Unless you run in debug mode

```
[ERROR] (file = TrackerDataFragment.icc)(func = DecodeModuleData)(line = 271)
| TrackerDataFragment::DecodeModuleData :: Module Data: ERROR code 0x1 for chip 9
```

- ❖ This is the decoding error that was flagged in the reconstruction.
- ❖ Still investigating as to what is the exact cause of this error.

000	Leader
aaaa	Chip address
eee	Error code
	001: No data available (The chip has not received an L1 command)
	010: Buffer Overflow
	100: Buffer Error (Soft Reset is needed.)
1	Separator bit

Table 14: Error data packet: 11 bits.

Figure: Tracker Readout Spec

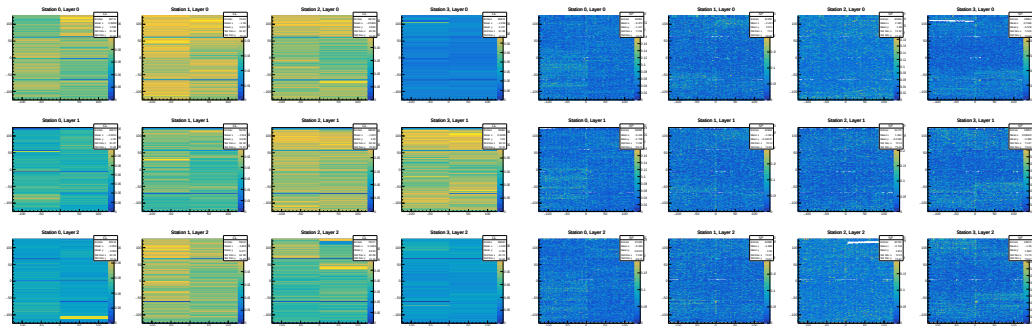
Is it recoverable?

- ❖ It is still unclear.
- ❖ A preliminary test was to comment out this error check in Calypso.

```
else if (sctEvent->HasError())  
{  
    // FIXME: Should we return failure here?  
    ATH_MSG_ERROR("Online module " << onlineModuleID << " reports one or more errors.");  
    // continue;  
    //Let's fill the data and see what happens  
}
```

- ❖ This seems to run **none of the other checks are triggered (? Pawan: Verify this)**
 - ❖ This is probably a good sign.
- ❖ This is not a proper fix, but just a test to see if data is recovered.
 - ❖ Still need to identify and fix the root cause of the error.
 - ❖ If not fixable, then we need to handle it correctly in the reconstruction.
- ❖ Some results in the next slides.

Cluster and Space Points after disabling the check



(a) Clusters from Run #022300

(b) Space Points from Run #022300

Figure: SCT Clusters and Space Points from Run #022300 after disabling the error check mentioned above.

SCT Hits

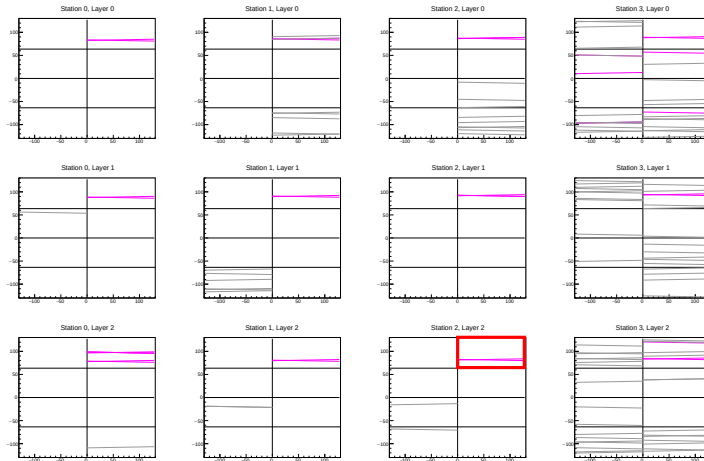


Figure: SCT Hits from Event #2840829, Run #022300 after disabling the error check mentioned above. Hits are restored(?) in the highlighted module.

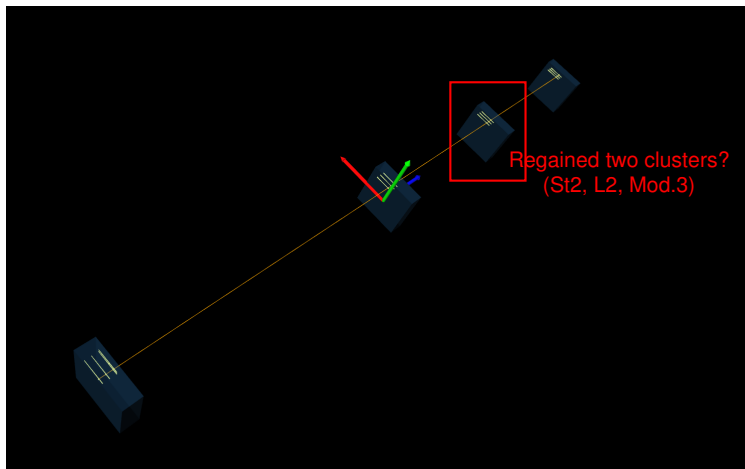


Figure: Event #2840829, Run #022300 after disabling the error check mentioned above.

Summary

- ❖ Some issues was observed in the 2026 Tracker data.
 - ❖ Specifically in St2, L2, Mod.3 didn't seem to have any hits/clusters
 - ❖ This could probably be isolated to a specific chip
 - ❖ Studies still ongoing to better understand this.
- ❖ The full extent of the impact on tracking is unclear
 - ❖ Probably not too huge
 - ❖ Could use some studies to quantify this.
- ❖ We could use some better error handling in the reco. side
- ❖ Partly recovered by disabling some checks.
 - ❖ This is good which means we can recover at least some of the data if not all.